# Data Analysis

*Deep Statistical Analysis for Research*

R Program + IBM SPSS Statistics | Practical academic workshop

# Table of Contents


- Chapter 1 - Objectives and research mindset
- Chapter 2 - Study design and data preparation
- Chapter 3 - Choosing the correct statistical method
- Chapter 4 - R program: step-by-step analysis
- Chapter 5 - SPSS: menu and syntax workflow
- Chapter 6 - Advanced methods and best tools
- Chapter 7 - Reporting, publishing and researcher path

# By the End, Participants Can

### 1 Plan

Turn a research question into an analysis plan.

### 2 Prepare

Clean data and build a reproducible workflow.

### 3 Analyze

Choose and run the right method in R and SPSS.

### 4 Report

Write results with estimates, CI, and interpretation.

CHAPTER ONE

# Research Mindset

*Statistics begins with better questions, not software.*

# Statistics Is a Research Skill

- A strong analysis starts before data collection.
- The study design decides what you can claim.
- Software gives output; the researcher gives meaning.
- Good statistics makes decisions visible and reproducible.

CHAPTER TWO

# Study Design and Data Preparation

*Build the analysis plan before opening R or SPSS.*

# From Research Problem to Analysis Plan

| 1 Problem | 2 Gap | 3 Question | 4 Variables | 5 Method | 6 Report |
|---|---|---|---|---|---|
| What matters? | Define clearly | Define clearly | Define clearly | Define clearly | What changed? |

**If this chain is weak, the analysis becomes weak.**

# Study Design Determines the Claim

### 1 Cross-sectional

Association, prevalence, snapshot.

### 2 Case-control

Rare outcomes, odds ratios.

### 3 Cohort

Risk, incidence, temporality.

### 4 Trial

Causal effect when done well.

# Data Must Be Analysis-Ready

- One row = one participant or observation.
- Each variable has a name, label, type, and unit.
- Missing values are coded intentionally.
- Cleaning decisions are saved in syntax or script.

CHAPTER THREE

# Choosing the Correct Statistical Method

*Use the outcome, design, and objective to choose the method.*

# The 3 Questions Before Any Test

### 1 Outcome

Continuous, binary, ordinal, count, or time-to-event?

### 2 Design

Independent, paired, clustered, or repeated?

### 3 Goal

Describe, compare, associate, adjust, or predict?

**Answer these first. Then choose the test.**

# Common Method Map

### 2 independent means
Welch t-test

### 3+ means
ANOVA / Welch ANOVA

### Categories
Chi-square / Fisher

### Adjusted binary outcome
Logistic regression

### Time-to-event
Kaplan-Meier / Cox

### Repeated data
Mixed model / GEE

# Do Not Worship the P-Value

### 1 Estimate

What changed?

### 2 95% CI

How precise is it?

### 3 P-value

How compatible with null?

### 4 Context

Does it matter clinically?

# Assumptions Are Diagnostic Questions

- Are observations independent?
- Is the outcome distribution reasonable?
- Are variances similar?
- Are influential cases driving the result?
- Does the model fit the research question?

CHAPTER FOUR

# R Program Workflow

*Reproducible analysis from raw data to report.*

# R Project Setup

```
install.packages(c("tidyverse", "janitor",
                   "gtsummary", "broom"))

library(tidyverse)
library(janitor)
library(gtsummary)
library(broom)
```

1 **Project**

Keep files organized.

2 **Script**

Save every step.

3 **Output**

Reproduce tables.

# R: Clean and Label Data

```
data <- raw |> clean_names() |>
  mutate(
    sex = factor(sex,
      labels = c("Male", "Female")),
    outcome = factor(outcome,
      labels = c("No", "Yes"))
  ) |>
  filter(age_years >= 18)
```

1

### Clean names

Avoid typing errors.

2

### Label codes

Make output readable.

3

### Filter rules

Document eligibility.

# R: Table and Plot

```
data |>
  select(age_years, sex, treatment, outcome) |>
  tbl_summary(by = treatment) |>
  add_p()

ggplot(data, aes(treatment, age_years)) +
  geom_boxplot() +
  theme_minimal()
```

1 **Table 1**

Describe sample.

2 **Plot**

Check distribution.

3 **Then test**

Do not skip inspection.

# R: Tests and Regression

```
t.test(score ~ treatment, data = data)
chisq.test(table(data$treatment, data$outcome))

fit <- glm(outcome ~ treatment + age_years + sex,
           data = data, family = binomial)

tidy(fit, exponentiate = TRUE,
     conf.int = TRUE)
```

1

### Compare

t-test / chi-square.

2

### Adjust

Regression model.

3

### Report

OR, CI, p-value.

PROGRAM IMAGE

# How to Use R During the Session

## RStudio-style workflow

### Script

```
data <- read_csv("study.csv") |> clean_names()
tbl_summary(data, by = treatment) |> add_p()
fit <- glm(outcome ~ treatment + age + sex,
family = binomial, data = data)
tidy(fit, exponentiate = TRUE, conf.int = TRUE)
```

### Environment

data: 150 observations, 12 variables

fit: logistic regression model

### Plots

### Console / Output

```
OR = 1.86 95% CI 1.12 to 3.09 p = 0.016
```

Interpretation: treatment is associated with higher odds of the outcome after adjustment.

1. **Script**
   Write every cleaning and analysis step.

2. **Run**
   Send code to the console and read messages.

3. **Inspect**
   Check tables, plots, and model output.

4. **Export**
   Save final tables and figures for manuscript.

PROGRAM OUTPUT

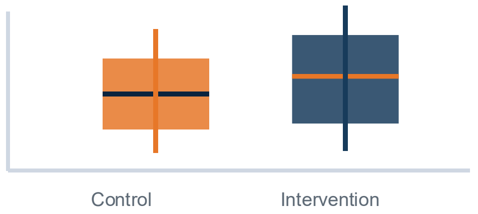
# R Output: Read the Result Easily

**R output: table, plot, model**

### Table 1 by treatment

| Variable | Control | Intervention |
|---|---|---|
| Age, mean (SD) | 42 (12) | 45 (11) |
| Female, n (%) | 31 (52%) | 34 (57%) |
| Outcome, n (%) | 8 (13%) | 15 (25%) |

### Boxplot check




### Regression output

```
term OR 95% CI p-value
treatment 1.86 1.12, 3.09 0.016
age_years 1.03 1.00, 1.06 0.041
```

1 **Table**

Describes the sample.

2 **Plot**

Shows distribution and outliers.

3 **Model**

Gives adjusted estimate.

4 **Sentence**

Translate output into meaning.

# Complete Data Analysis Workflow

1 **Ask**
Before analysis

2 **Design**
Before analysis

3 **Collect**
Before analysis

4 **Clean**
Before analysis

5 **Describe**
During analysis

6 **Model**
During analysis

7 **Validate**
During analysis

8 **Report**
During analysis

**Data analysis is a full workflow, not one statistical test.**

# What Data Analysis Includes

### Descriptive

Who is in the sample?

### Exploratory

What patterns or problems appear?

### Inferential

Is the observed difference compatible with chance?

### Modeling

What is the adjusted association?

### Prediction

How well can we predict future outcomes?

### Sensitivity

Does the conclusion survive reasonable changes?

# Minimum Analysis Plan

- Primary outcome and exposure
- Covariates and confounders
- Descriptive table and planned figures
- Main statistical test or model
- Missing-data and sensitivity approach
- Exact reporting format

CHAPTER FIVE

# SPSS Workflow

*Accessible analysis with saved syntax.*

# SPSS: Think Menu + Syntax

### 1 Data View

Rows are cases. Columns are variables.

### 2 Variable View

Set labels, values, missing, and measure.

### 3 Menus

Good for learning and checking options.

### 4 Syntax

Essential for reproducibility.

PROGRAM IMAGE

# SPSS Data View: Enter and Check Data

SPSS Data View

| id | age | sex | treatment | score | outcome | followup |
|---|---|---|---|---|---|---|
| 1001 | 24 | 1 | 0 | 72 | 1 | 30 |
| 1002 | 29 | 2 | 1 | 69 | 0 | 42 |
| 1003 | 34 | 1 | 0 | 66 | 0 | 54 |
| 1004 | 39 | 2 | 1 | 63 | 1 | 66 |
| 1005 | 44 | 1 | 0 | 60 | 0 | 78 |
| 1006 | 49 | 2 | 1 | 57 | 0 | 90 |
| 1007 | 54 | 1 | 0 | 54 | 1 | 102 |
| 1008 | 59 | 2 | 1 | 51 | 0 | 114 |

Data View | Variable View

1 **Rows**

Each row is one case.

2 **Columns**

Each column is one variable.

3 **Codes**

Values must match the codebook.

4 **Check**

Look for impossible values.

# SPSS Variable View: Make Output Readable

**SPSS Variable View**

| Name | Type | Label | Values | Measure |
|---|---|---|---|---|
| age | Numeric | Age in years | | Scale |
| sex | Numeric | Participant sex | 1 Male; 2 Female | Nominal |
| treatment | Numeric | Study group | 0 Control; 1 Intervention | Nominal |
| score | Numeric | Functional score | | Scale |
| outcome | Numeric | 30-day outcome | 0 No; 1 Yes | Nominal |
| followup | Numeric | Follow-up days | | Scale |

Data View | Variable View

1 **Name**

Short variable name.

2 **Label**

Full meaning.

3 **Values**

0/1 and category labels.

4 **Measure**

Scale, ordinal, nominal.

# SPSS: Import and Label

```
GET DATA /TYPE=XLSX
 /FILE='study.xlsx'
 /READNAMES=on.

VARIABLE LABELS
 age_years 'Age in years'
 outcome 'Study outcome'.

VALUE LABELS outcome
 0 'No' 1 'Yes'.
EXECUTE.
```

1 **Import**

Read data safely.

2 **Label**

Make output clear.

3 **Save**

Keep .sps file.

# SPSS: Main Menu Paths

### Descriptives

Analyze > Descriptive Statistics

### t-test / ANOVA

Analyze > Compare Means

### Chi-square

Analyze > Descriptive Statistics > Crosstabs

### Regression

Analyze > Regression

### Survival

Analyze > Survival

# SPSS: Regression Syntax

```
REGRESSION
 /DEPENDENT score
 /METHOD=ENTER treatment age_years sex.

LOGISTIC REGRESSION VARIABLES outcome
 /METHOD=ENTER treatment age_years sex
 /PRINT=CI(95).
```

1 **Linear**

Continuous outcome.

2 **Logistic**

Binary outcome.

3 **Interpret**

B or Exp(B) + CI.

# SPSS Output: Read the Viewer

## SPSS Output Viewer style

### Output Navigator

- Frequencies
  - Statistics
  - Frequency Table
- Crosstabs
  - Case Processing
  - Chi-Square Tests
- Logistic Regression
  - Variables in Equation
  - Classification Table

### Variables in the Equation

| Variable | B | Sig. | Exp(B) | 95% C.I. |
|---|---|---|---|---|
| Treatment | 0.62 | .016 | 1.86 | 1.12 - 3.09 |
| Age | 0.03 | .041 | 1.03 | 1.00 - 1.06 |

**How to read it**

Exp(B) is the odds ratio. Sig. is the p-value. The CI shows precision.

1 **Navigator**

Find each analysis.

2 **Table**

Read the correct row.

3 **Exp(B)**

Odds ratio.

4 **CI**

Precision of estimate.

CHAPTER SIX

# Advanced Methods and Best Tools

*Deep topics without overwhelming the slide.*

# Advanced Methods You Should Know

### Sample size

Power and clinically meaningful effect.

### Missing data

Diagnose pattern before deleting.

### Survival

Time-to-event and censoring.

### Reliability

Questionnaires and scale quality.

### Prediction

Validation, calibration, discrimination.

### Meta-analysis

Combine evidence across studies.

# Missing Data: The Safe Approach

- Measure how much is missing.
- Check whether missingness differs by group.
- Avoid silent complete-case deletion.
- Use imputation or sensitivity analysis when needed.

# Survival Analysis: When Time Matters

### 1 Need

Follow-up time + event status.

### 2 Describe

Kaplan-Meier curve.

### 3 Compare

Log-rank test.

### 4 Adjust

Cox regression.

TOOLKIT

# Best Tools for Research Analysis

### REDCap
Data capture

### R + SPSS
Analysis

### Quarto
Reports

### Zotero
References

### Rayyan / RevMan
Reviews

### OSF / GitHub
Transparency

CHAPTER SEVEN

# Reporting and Researcher Path

*Turn analysis into publishable evidence.*

# How to Report a Result

**Treatment was associated with lower systolic BP by 7.8 mmHg (95% CI 2.1 to 13.4 lower; p = 0.008).**

- Direction: lower or higher?
- Size: how much?
- Uncertainty: 95% CI
- Meaning: clinical and research context

# Tables and Figures That Look Academic

1
### Table 1
Sample description.

2
### Table 2
Main adjusted result.

3
### Figure 1
Flow, distribution, survival, or forest plot.

4
### Footnotes
Tests, reference groups, missing data.

# Avoid These Common Traps

- Choosing tests after searching for significance.
- Ignoring missing data or denominators.
- Calling association causation.
- Overfitting small datasets.
- Reporting only p-values.

# From Student to Research Leader

| | | | | |
|---|---|---|---|---|
| 1 **Learn** | 2 **Analyze** | 3 **Publish** | 4 **Collaborate** | 5 **Mentor** |
| Methods | R/SPSS | Papers | Networks | Others |

**You just have to start, then keep improving the quality of your evidence.**

# Final Checklist

- Question is clear.
- Data are clean.
- Method matches design.
- Assumptions are checked.
- Result includes estimate + CI.
- Interpretation is honest.

# Key References and Tools

- R Project: r-project.org
- IBM SPSS Statistics documentation
- STROBE and EQUATOR reporting guidelines
- Quarto, REDCap, Zotero, Rayyan, RevMan